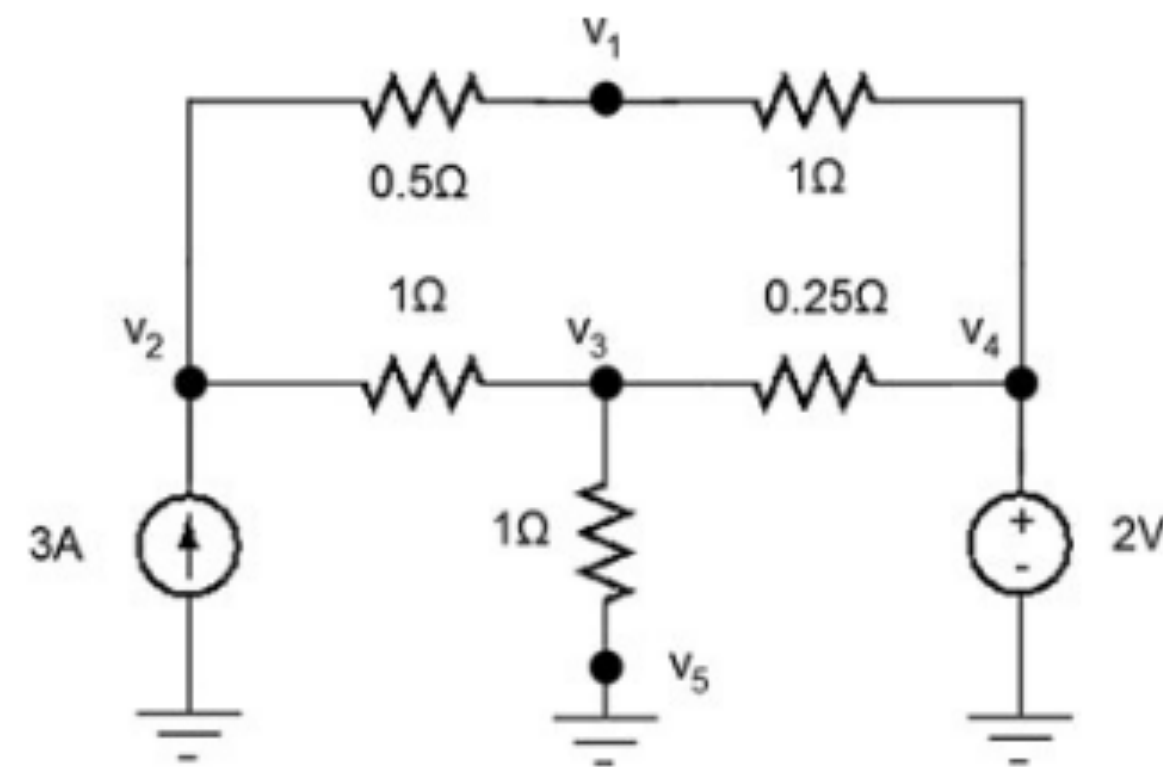
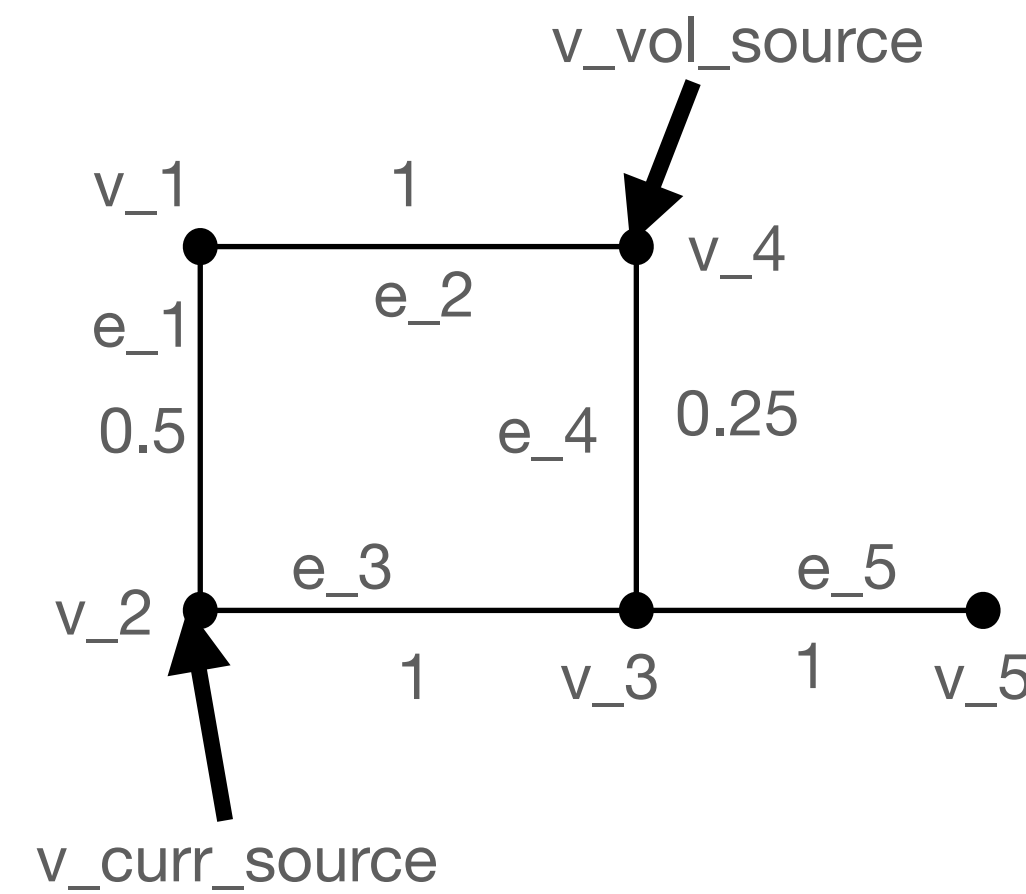


A Circuit toy example



(Grady et al. 2010)

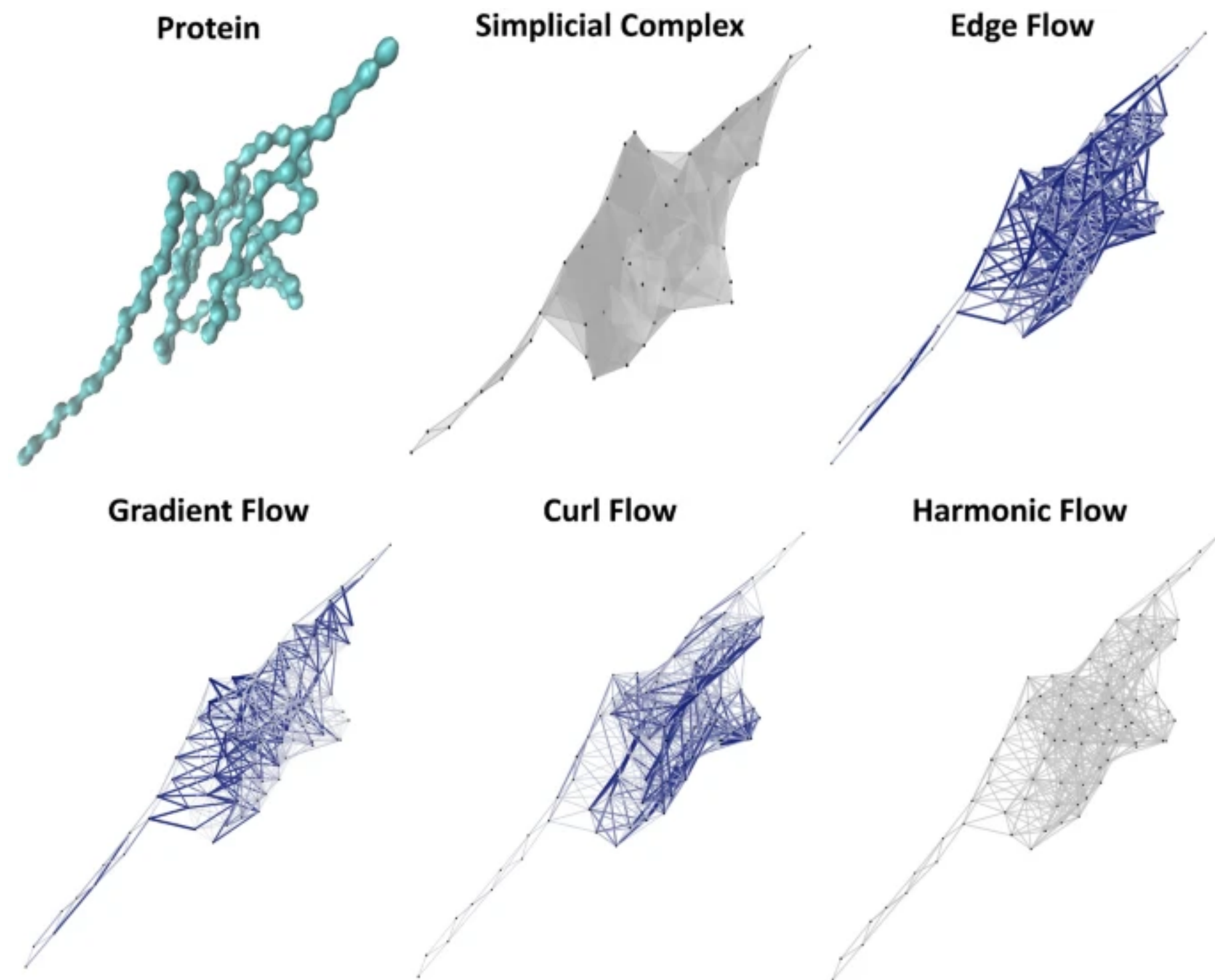


Voltage (Kirchhoff's voltage law)
gradient flow, curl-free

$$\mathbf{B}_1 = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 1 \\ 0 & -1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{pmatrix} \quad \mathbf{v}_{vol} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ 2 \\ 0 \end{pmatrix}$$

Current (Kirchhoff' current law)
curl flow, div-free

Applications of Hodge decomposition



Other concerned applications:

- **Biomolecular structure analysis** (*Chen et al. 2021; Wei et al. 2022*)
- RNA velocity estimation (*chen et al. 2023*)
- Brain networks (*Anand et al. 2022*)
- **Force fields in atomic systems?**